

MUSHARAF KHAN PATHAN

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Software Engineer with production full-stack and cloud experience across Java, Python, and TypeScript. Shipped 35+ REST APIs and cut deployment time from 4 hours to 15 minutes with Docker, Kubernetes, and GitLab CI/CD on AWS; co-founded a product serving 42 weekly active users at sub-1% failure rate and 1.8s p50 latency. Owns features end to end, from object-oriented design and unit/integration testing through deployment, on-call debugging, and root-cause analysis on agile cross-functional teams.

WORK EXPERIENCE

Ocreda (Co-Founded Startup)

06/2026 - Present

Co-Founder and Full-Stack Engineer

- Scaled a production TypeScript, Next.js, and React application on Supabase and PostgreSQL to 42 weekly active users and 1,300+ records at 38% week-over-week retention, owning design through deployment on a 6-person agile team.
- Cut invalid data writes to zero by building a validation pipeline in Deno Edge Functions that enforces strict JSON schemas and verifies record IDs against PostgreSQL before commit, rejecting malformed LLM output and hallucinated references.
- Hold 1.8s p50 and 3.4s p95 retrieval latency at a sub-1% failure rate by monitoring service health, reading logs and dashboards, and shipping fixes through pull-request review and automated test pipelines.

Colorado Resilience

05/2025 - 05/2026

Software Engineer Intern

- Built and shipped 35+ production REST API endpoints with Flask, SQLAlchemy, and OpenAPI 3.0, plus reusable React and Material UI components, for a case-management platform used by state agency staff and the public.
- Reduced release deployment time from 4 hours to 15 minutes by containerizing microservices with Docker and automating GitLab CI/CD pipelines deploying to Kubernetes on AWS EKS with zero-downtime rolling updates.
- Restored access for all blocked users by tracing an authentication failure across browser tokens, Flask logs, Keycloak configuration, and PostgreSQL role records, then fixing the claim mapping and adding regression tests.
- Improved median page-load time about 40% across the five most-used workflows via request tracing, Redux caching, and render optimization; wrote unit, integration, and end-to-end tests in agile scrum.
- Onboarded 3 new engineers in under a week by authoring environment setup, deployment, and troubleshooting documentation covering Docker, AWS, Git, and database configuration.

Illinois Institute of Technology

01/2024 - 12/2025

Undergraduate Teaching Assistant

- Led 20+ lab and debugging sessions for 100+ students across 3 courses in object-oriented Java and systems programming (C, x86 Assembly), covering OO design, data structures, GDB debugging, and Unix process control.
- Evaluated 500+ assignments using Valgrind memory-leak analysis and algorithmic-complexity feedback, reinforcing coding standards, testing discipline, and root-cause analysis.

EDUCATION

Bachelor of Science, Computer Science

01/2022 - 05/2026

Illinois Institute of Technology, Chicago, IL • Minor in Architecture • GPA: 3.52/4.00 • Cum Laude • Dean's List (5x)

Relevant Coursework – Data Structures and Algorithms, Introduction to Algorithms, Operating Systems, Systems Programming, Databases, Computer Organization and Assembly Language, Programming Languages and Translators, Software Engineering, Machine Learning, Information Security, Object-Oriented Programming I and II.

PROJECTS

Machine Unlearning Research (PyTorch)

01/2026 - 05/2026

Benchmarked 6 unlearning methods across 6 metrics on ResNet-18 and CIFAR-10 (92.35% baseline accuracy) under 3 stress tests; built Retain-Protected Gradient Ascent reaching 0% forget accuracy at 80.3% retain accuracy, cutting membership-inference privacy leakage 17x.

CheckURL, Phishing Scanner Browser Extension (JavaScript)

08/2025 - 11/2025

Shipped a pre-click URL scanner achieving 92% detection accuracy and an 84% reduction in dangerous clicks against the URLhaus malicious dataset at sub-900ms scan time, with defensive parsing and security-focused UX.

SKILLS

Core: Java, Python, C, C++, JavaScript, TypeScript, SQL, Bash, object-oriented design, data structures and algorithms

Application Development: React, Next.js, Node.js, Spring Boot, Flask, FastAPI, REST APIs, Swagger/OpenAPI, microservices

Cloud & DevOps: AWS (EC2, S3, RDS, EKS), Docker, Kubernetes, Helm, Argo CD, GitLab CI/CD, Git, Linux/Unix

Databases: PostgreSQL, MySQL, Oracle SQL, SQLite, Redis

Testing & Quality: pytest, JUnit, unit/integration testing, test automation, GDB, Valgrind, monitoring, root-cause analysis, agile/scrum

AI & Automation: PyTorch, scikit-learn, LLM integration and output validation, Claude Code, GitHub Copilot